Transfer-strategy results: thesis tables and figures

October 30, 2026

Table 1: Combination of TimesFM (ctx 1024), Chronos-2 (ctx 8192) and TabPFN-TS: aggregate relative MASE (↓), change vs TimesFM with 95% interval, and regret vs the ex-post best member per task. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only. Intervals: 95% task-stratified bootstrap (2,000 replicates) resampling series and windows.

Method	Score	% vs TimesFM [CI]	Mean regret	Max regret
NNLS, per task	0.633	−4.5 [−6.6, −2.5]	1.002	1.025
Inverse MASE^4, per series	0.633	−4.5 [−6.2, −2.7]	1.002	1.041
NNLS, per task × 3 horizon bins	0.634	−4.3 [−6.4, −2.3]	1.004	1.015
NNLS, per task × 6 horizon bins	0.635	−4.1 [−6.3, −2.1]	1.006	1.021
Inverse MASE^2, top 2, per series	0.637	−3.9 [−6.0, −2.0]	1.008	1.042
NNLS, per series	0.638	−3.7 [−5.8, −1.6]	1.011	1.052
Best member, per task	0.640	−3.5 [−5.6, −1.5]	1.013	1.063
Best member, per series	0.642	−3.1 [−5.2, −1.2]	1.016	1.052
Inverse MASE^2, per series	0.644	−2.9 [−5.3, −0.4]	1.020	1.136
NNLS, per task × 3 bins, 50% shrinkage	0.652	−1.6 [−4.6, +2.0]	1.037	1.346
Best member, per series, last window	0.653	−1.5 [−3.9, +0.9]	1.034	1.111
NNLS, global	0.657	−0.9 [−2.2, +1.2]	1.041	1.089
Inverse MASE, per series	0.660	−0.4 [−3.2, +3.0]	1.052	1.451
FFORMA-style meta-learner (LODO)	0.661	−0.3 [−2.5, +3.3]	1.046	1.094
TimesFM	0.663	–	1.050	1.101
Equal mean	0.682	+2.9 [−0.5, +7.5]	1.102	1.939
Median	0.684	+3.2 [−0.3, +7.8]	1.120	2.256
Chronos-2	0.702	+5.9 [+1.4, +11.1]	1.181	2.899
TabPFN-TS	0.773	+16.6 [+11.1, +24.1]	1.287	2.919

Table 2: Residuals $e = (\hat{y} - y)/s$ of the zero-shot models at every forecast point (s: the series' MASE scale). Pairwise Pearson correlation of residuals, pooled over the task's points (all) and the median over series of within-series correlations (ser.); mean residual (bias; positive: over-forecasting). Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	TFM–Chr all	ser.	TFM–TabPFN all	ser.	Chr–TabPFN all	ser.	Bias TFM	Bias Chr	Bias TabPFN
M4 Hourly	0.78	0.85	0.77	0.79	0.75	0.79	−0.14	−0.13	−0.15
ETTh1/48	0.91	0.94	0.88	0.89	0.93	0.94	+0.11	+0.01	+0.01
ETTh1/96	0.92	0.94	0.89	0.90	0.92	0.92	+0.01	−0.05	−0.09
ETTh2/48	0.94	0.93	0.87	0.91	0.89	0.93	+0.09	+0.10	+0.14
ETTm1/48	0.91	0.91	0.82	0.85	0.77	0.85	−0.09	−0.18	−0.08
ETTm2/48	0.91	0.95	0.84	0.83	0.83	0.86	−0.10	−0.08	−0.21
ECL	0.90	0.94	0.87	0.90	0.82	0.85	−0.03	−0.02	−0.13
Traffic	0.93	0.93	0.84	0.84	0.86	0.85	+0.00	−0.01	−0.09
Weather	0.93	0.89	0.92	0.85	0.89	0.83	−0.05	−0.02	−0.08
Exchange	0.98	0.98	0.85	0.85	0.83	0.82	+2.10	+1.42	+3.00
ILI	0.07	−0.10	0.14	−0.07	0.83	0.88	+0.15	−0.83	−0.17
Median	0.91	0.93	0.85	0.85	0.83	0.85			

Table 3: Error redundancy of the zero-shot foundation models (squared scaled errors). Shared: share of member error explained by the members' mean error; 1st PC: share of the first principal component. Change in MSE vs the best member for ex-post optimal simplex weights (ideal), equal weights, and ideal weights with SeasonalNaive, AutoETS and AutoTheta added (+stats). TimesFM–Chronos-2 pair: uncentred error correlation ρ, accuracy ratio $k = \sigma_1/\sigma_2$ and the Bates–Granger optimum $(1-\rho^2)/(1+k^2-2\rho k)$, which is below one only if $\rho < k$. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	Shared	1st PC	Best	Ideal	Equal	Ideal +stats	ρ	k	Pair opt.
M4 Hourly	0.85	0.84	TimesFM	−8.2	−7.2	−8.4	0.79	0.93	−5.1
ETTh1/48	0.94	0.94	TabPFN-TS	−3.3	−2.7	−6.2	0.90	0.97	−2.4
ETTh1/96	0.94	0.94	TabPFN-TS	−3.1	−1.4	−6.9	0.92	0.99	−3.3
ETTh2/48	0.93	0.93	Chronos-2	−1.2	+1.5	−1.4	0.94	0.94	0.0
ETTm1/48	0.89	0.89	Chronos-2	−11.6	−4.3	−14.7	0.91	0.90	0.0
ETTm2/48	0.90	0.90	Chronos-2	−1.9	+2.4	−6.2	0.91	0.97	−1.8
ECL	0.90	0.91	Chronos-2	−4.3	+1.1	−4.3	0.90	0.99	−4.3
Traffic	0.91	0.92	TimesFM	−0.4	+7.9	−0.9	0.93	0.95	−0.4
Weather	0.94	0.94	Chronos-2	−2.4	−1.0	−15.3	0.93	0.96	−1.0
Exchange	0.92	0.92	Chronos-2	0.0	+9.3	−0.2	0.97	0.96	−0.2
ILI	0.60	0.86	TimesFM	−12.2	+206.1	−12.6	0.01	0.38	−12.2
Median	0.91	0.92		−3.1	+1.1	−6.2	0.91	0.96	−1.8

Table 4: TimesFM context length: change in relative MASE vs ctx 1024 (%, ↓), flip invariance on. First column: relative MASE at ctx 1024. No flip: ctx 16256 without flip invariance. Selected: context and flip setting chosen per task from earlier windows (12 variants). Chronos-2: ctx 8192, for reference. Bold: best context per task. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	1024 (rel.)	512	1024	2048	4096	8192	16256	No flip	Selected	Chronos-2
M4 Hourly	0.578	+0.9	0	0.0	0.0	0.0	0.0	+3.2	0.0	+9.7
ETTh1/48	1.018	−0.8	0	−0.7	**−1.5**	−0.3	−1.1	+1.1	+0.7	−6.9
ETTh1/96	0.912	+0.5	0	−0.9	−1.3	**−1.4**	−1.3	−0.1	+1.2	−2.1
ETTh2/48	0.885	−2.4	0	−4.8	−4.6	**−4.9**	−4.6	−6.0	−5.0	−8.7
ETTm1/48	0.749	**−10.7**	0	+1.5	−0.7	−0.4	−0.2	−2.1	−6.1	−9.2
ETTm2/48	0.680	+1.6	0	+2.2	**−2.9**	−1.7	−1.5	−2.1	−2.4	−5.4
ECL	0.704	+0.7	0	+0.4	−0.8	+0.5	**−3.9**	−1.3	−3.7	−6.2
Traffic	0.523	+11.8	0	−5.4	−8.0	−10.8	**−11.5**	−9.7	−11.5	−0.5
Weather	0.451	+3.3	0	−2.9	−7.4	**−11.5**	−10.2	−7.8	−4.8	−8.9
Exchange	0.924	+1.2	0	−1.0	−3.5	−3.9	**−3.9**	−2.9	−2.6	−2.7
ILI	0.292	+13.1	0	0.0	0.0	0.0	0.0	+1.5	+0.3	+189.9
Aggregate	0.663	+1.6	0	−1.1	−2.8	−3.2	−3.5	−2.5	−3.2	+5.9

Table 5: Effect of turning flip invariance off: change in TimesFM relative MASE (%) vs flip on at the same context. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	512	1024	2048	4096	8192	16256
M4 Hourly	+2.8	+3.2	+3.2	+3.2	+3.2	+3.2
ETTh1/48	+1.2	−1.7	+1.6	+3.0	+2.8	+2.2
ETTh1/96	+1.7	+0.3	+1.9	+1.8	+1.3	+1.2
ETTh2/48	+0.8	−0.5	−2.1	−1.5	−0.4	−1.5
ETTm1/48	+1.9	−1.7	−0.1	+0.2	−0.7	−1.9
ETTm2/48	+1.7	+2.9	+0.8	−1.0	+0.4	−0.7
ECL	+2.6	+2.9	+4.6	+3.8	+4.1	+2.7
Traffic	+0.6	+1.9	+3.0	+2.2	+1.9	+2.0
Weather	+1.6	+1.9	+2.6	+0.8	+1.5	+2.7
Exchange	+0.3	+1.7	+0.6	−0.1	+1.0	+1.0
ILI	+7.2	+1.5	+1.5	+1.5	+1.5	+1.5

Table 6: Chronos-2 relative MASE (↓) by context length. Evaluated windows 0–16 (0–6 for M4 and ILI); weights and selections use earlier windows only.

Task	1024	2048	4096	8192
M4 Hourly	**0.634**	**0.634**	**0.634**	**0.634**
ETTh1/48	0.954	0.945	**0.944**	0.948
ETTh1/96	0.903	0.896	0.895	**0.893**
ETTh2/48	0.852	0.810	0.810	**0.808**
ETTm1/48	**0.662**	0.695	0.683	0.680
ETTm2/48	0.665	0.659	0.645	**0.643**
ECL	**0.656**	0.658	0.659	0.660
Traffic	**0.514**	0.515	0.517	0.521
Weather	0.458	0.418	0.414	**0.411**
Exchange	0.937	0.929	0.910	**0.900**
ILI	**0.845**	**0.845**	**0.845**	**0.845**
Aggregate	0.715	0.707	0.703	0.702

Table 7: LoRA fine-tuning per task: change in relative MASE (%, ↓) vs the zero-shot model at the same inference context. TimesFM: LoRA trained at ctx 1024 (r=4, all linear layers), evaluated at ctx 1024 (vs zero-shot 1024) and at ctx 16256 (vs zero-shot 16256) for seeds 0–2. Chronos-2: library-default LoRA trained at ctx 2048, evaluated at 2048 and 8192. Score columns: zero-shot relative MASE. *: validation selection kept the zero-shot model. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Task	TFM 16256	LoRA@1024	LoRA@16256 s0	s1	s2	Chr 8192	LoRA@2048	LoRA@8192
M4 Hourly	0.584	−2.5	−2.5	−1.8	−3.1	0.650	−0.4	−0.4
ETTh1/48	1.029	−5.9	−4.0	0.0*	0.0*	0.943	−0.8	+0.3
ETTh1/96	0.917	−1.7	−0.4	−2.3	−2.6	0.916	−2.0	−0.4
ETTh2/48	0.847	−3.4	−0.2	+0.2	−0.8	0.830	0.0*	0.0*
ETTm1/48	0.729	0.0*	0.0*	0.0*	−2.6	0.687	0.0*	0.0*
ETTm2/48	0.653	0.0*	0.0*	0.0*	0.0*	0.628	0.0*	0.0*
ECL	0.675	−8.2	−4.4	−4.2	−4.9	0.658	−0.2	−0.2
Traffic	0.481	−1.5	+0.5	+1.1	+0.1	0.539	−0.5	−0.2
Weather	0.386	−0.2	+1.1	+1.0	+0.6	0.393	+1.9	+0.8
Exchange	0.862	−2.7	+1.0	−1.4	+0.2	0.875	0.0*	0.0*
ILI	0.260	0.0*	0.0*	0.0*	0.0*	0.838	0.0*	0.0*
Aggregate	0.630	−2.4	−0.8	−0.7	−1.2	0.702	−0.2	0.0

Table 8: Joint comparison of transfer strategies: aggregate relative MASE (↓), change vs default TimesFM and vs TimesFM at ctx 16256 (% [95% CI]), GPU seconds per forecast origin summed over the 11 tasks, and one-time training GPU minutes (RTX A5000; TimesFM context capped at the longest history; adapters merged; TabPFN-TS from run logs). Combinations: task-level NNLS weights fitted on earlier windows; their cost is the sum of the members'. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring. Intervals: 95% task-stratified bootstrap (2,000 replicates) resampling series and windows.

Strategy	Score	% vs default [CI]	% vs ctx 16256 [CI]	GPU s/origin	Train min
TimesFM, ctx 1024 (default)	0.655	–	+3.9 [+1.1, +6.4]	5.8	–
TimesFM, ctx 8192	0.633	−3.4 [−5.5, −1.0]	+0.4 [−0.4, +1.4]	37.4	–
TimesFM, ctx 16256	0.630	−3.8 [−6.0, −1.1]	–	72.7	–
TimesFM, context selected per task	0.630	−3.9 [−6.6, −1.5]	−0.1 [−2.0, +1.7]	197.1	–
TimesFM LoRA, inference ctx 1024	0.640	−2.4 [−4.0, −1.2]	+1.5 [−1.1, +3.8]	5.9	63
TimesFM LoRA, inference ctx 16256 (seed 0)	0.625	−4.6 [−7.1, −1.8]	−0.8 [−1.8, +0.2]	72.7	63
TimesFM LoRA, inference ctx 16256 (seed 1)	0.626	−4.5 [−7.0, −1.8]	−0.7 [−1.7, +0.2]	72.7	63
TimesFM LoRA, inference ctx 16256 (seed 2)	0.623	−5.0 [−7.5, −2.3]	−1.2 [−2.2, −0.3]	72.7	63
Chronos-2, ctx 8192	0.702	+7.2 [+2.5, +13.7]	+11.4 [+6.5, +18.0]	17.1	–
Chronos-2 LoRA, inference ctx 8192	0.702	+7.2 [+2.4, +13.7]	+11.4 [+6.5, +17.7]	18.2	175
TabPFN-TS	0.758	+15.7 [+9.4, +24.4]	+20.3 [+13.4, +29.0]	4079.5	–
TimesFM 1024 + Chronos-2 2048	0.633	−3.5 [−5.6, −1.5]	+0.4 [−2.4, +2.8]	10.5	–
TimesFM 1024 + Chronos-2 8192	0.629	−4.0 [−6.2, −1.8]	−0.2 [−2.9, +2.2]	23.0	–
TimesFM 1024 + Chronos-2 8192 + TabPFN-TS	0.629	−4.0 [−6.6, −1.5]	−0.3 [−3.0, +2.2]	4102.5	–
TimesFM 16256 + Chronos-2 8192	0.619	−5.5 [−8.1, −2.9]	−1.8 [−3.8, 0.0]	89.8	–
TimesFM LoRA@16256 + Chronos-2 8192	0.617	−5.8 [−8.6, −3.1]	−2.1 [−4.1, −0.1]	89.8	63
TimesFM LoRA@16256 + Chronos-2 LoRA@8192	0.618	−5.7 [−8.6, −2.8]	−2.0 [−4.1, +0.1]	90.9	239

Table 9: Combination gain (combination vs its TimesFM member, %) and differences between gains (ratio of ratios, %; positive: the gain is smaller than in the reference). Intervals: series and windows resampled, or series and circular blocks of three consecutive windows. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Quantity	% [CI], series × windows	% [CI], series × window blocks
Combination gain, TimesFM default + Chronos-2	−4.0 [−6.2, −1.8]	−4.0 [−6.2, −2.0]
Combination gain, TimesFM 16256 + Chronos-2	−1.8 [−3.8, 0.0]	−1.8 [−3.6, −0.3]
Combination gain, TimesFM LoRA@16256 + Chronos-2	−1.3 [−3.2, +0.5]	−1.3 [−3.0, +0.1]
Combination gain, TimesFM LoRA@16256 + Chronos-2 LoRA	−1.2 [−3.1, +0.7]	−1.2 [−3.0, +0.3]
Difference: 16256 vs default	+2.3 [+0.3, +4.0]	+2.3 [+0.3, +4.2]
Difference: LoRA@16256 vs 16256	+0.5 [0.0, +1.1]	+0.5 [+0.1, +1.1]
Difference: LoRA@16256 vs default	+2.8 [+0.8, +4.7]	+2.8 [+0.7, +4.9]
Difference: Chronos-2 LoRA added	+0.1 [−0.4, +0.6]	+0.1 [−0.3, +0.6]

Table 10: Adding TabPFN-TS as a third member: aggregate relative MASE (↓) of the TimesFM + Chronos-2 pair and of the pair plus TabPFN-TS (task-level NNLS weights), and the change from adding TabPFN-TS (% [95% CI]). Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Combination	Pair	+ TabPFN-TS	Change, series × windows	Change, series × blocks
TimesFM ctx 1024 + Chronos-2	0.629	0.629	−0.1 [−1.1, +1.0]	−0.1 [−0.9, +0.9]
TimesFM ctx 16256 + Chronos-2	0.619	0.620	+0.1 [−0.7, +1.0]	+0.1 [−0.6, +1.0]
TimesFM LoRA@16256 + Chronos-2	0.617	0.618	+0.1 [−0.7, +1.0]	+0.1 [−0.6, +0.9]

Table 11: Relative MASE (↓) per task and strategy. Bold: best per task. TFM: TimesFM, ctx 1024 (default); TFM-L: TimesFM, ctx 16256; TFM-sel: TimesFM, context selected per task; TFM-LoRA: TimesFM LoRA, inference ctx 1024; TFM-L-LoRA: TimesFM LoRA, inference ctx 16256 (seed 0); Chr: Chronos-2, ctx 8192; Chr-LoRA: Chronos-2 LoRA, inference ctx 8192; TFM+Chr2k: TimesFM 1024 + Chronos-2 2048; TFM+Chr: TimesFM 1024 + Chronos-2 8192; TFM-L+Chr: TimesFM 16256 + Chronos-2 8192; TFM-L-LoRA+Chr: TimesFM LoRA@16256 + Chronos-2 8192. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Task	*TFM*	*TFM-L*	*TFM-sel*	*TFM-LoRA*	*TFM-L-LoRA*	*Chr*	*Chr-LoRA*	*TFM+Chr2k*	*TFM+Chr*	*TFM-L+Chr*	*TFM-L-LoRA+Chr*
M4 Hourly	0.584	0.584	0.584	0.569	0.569	0.650	0.648	0.571	0.571	0.571	**0.565**
ETTh1/48	1.044	1.029	1.050	0.983	0.988	**0.943**	0.946	0.983	0.982	0.989	0.973
ETTh1/96	0.924	0.917	0.926	**0.908**	0.913	0.916	0.913	0.918	0.914	0.918	0.914
ETTh2/48	0.886	0.847	0.853	0.856	0.845	0.830	0.830	**0.828**	0.831	0.830	0.831
ETTm1/48	0.743	0.729	0.689	0.743	0.729	**0.687**	**0.687**	0.722	0.708	0.720	0.720
ETTm2/48	0.658	0.653	0.644	0.658	0.653	**0.628**	**0.628**	0.638	0.629	0.629	0.629
ECL	0.706	0.675	0.684	0.648	**0.645**	0.658	0.656	0.671	0.666	0.658	0.646
Traffic	0.548	**0.481**	**0.481**	0.540	0.483	0.539	0.538	0.540	0.541	**0.481**	0.483
Weather	0.433	0.386	0.391	0.433	0.391	0.393	0.396	0.407	0.404	**0.385**	0.386
Exchange	0.899	**0.862**	0.862	0.875	0.871	0.875	0.875	0.895	0.890	**0.862**	0.871
ILI	0.260	0.260	0.260	0.260	0.260	0.838	0.838	0.250	0.250	**0.250**	0.250
Aggregate	0.655	0.630	0.630	0.640	0.625	0.702	0.702	0.633	0.629	0.619	0.617

Table 12: Per-task Diebold–Mariano tests on window-level MASE (Newey–West variance, HLN correction, Holm across the 11 tasks): tasks with lower mean loss / significantly lower / significantly higher at 5%. TimesFM ctx 16256 is a reference and is not tested against the default here. Common evaluation set: windows 0–13 (0–3 for M4 and ILI); the three oldest windows, used for fine-tuning early stopping, are excluded from all fitting and scoring.

Strategy	vs TimesFM default	vs TimesFM 16256
TimesFM, context selected per task	9 / 0 / 0	4 / 0 / 0
TimesFM LoRA, inference ctx 1024	10 / 1 / 0	5 / 0 / 0
TimesFM LoRA, inference ctx 16256 (seed 0)	11 / 0 / 0	6 / 1 / 0
Chronos-2, ctx 8192	9 / 0 / 1	6 / 0 / 1
Chronos-2 LoRA, inference ctx 8192	9 / 0 / 1	6 / 0 / 1
TimesFM 1024 + Chronos-2 2048	11 / 1 / 0	7 / 0 / 0
TimesFM 1024 + Chronos-2 8192	11 / 1 / 0	8 / 0 / 0
TimesFM 16256 + Chronos-2 8192	11 / 0 / 0	8 / 0 / 0
TimesFM LoRA@16256 + Chronos-2 8192	11 / 0 / 0	8 / 2 / 0

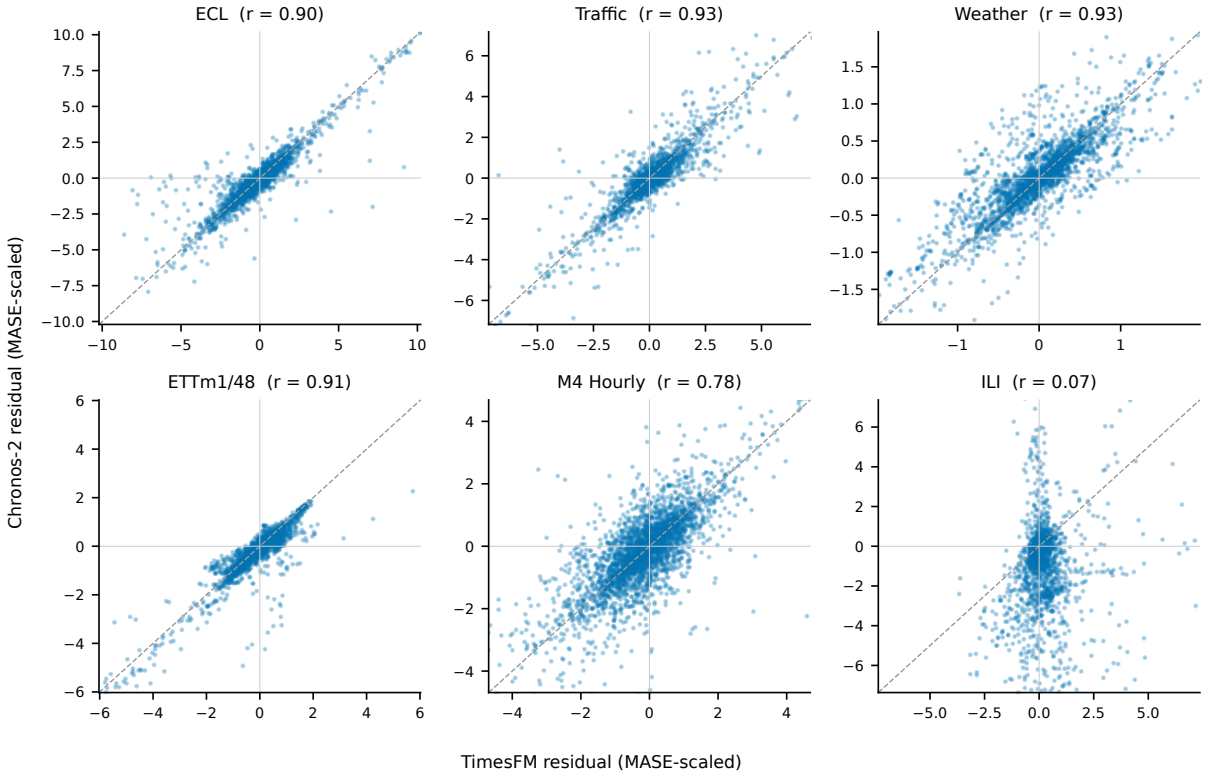


Figure 1: TimesFM vs Chronos-2 residuals at individual forecast points (random sample of 4,000 per task).

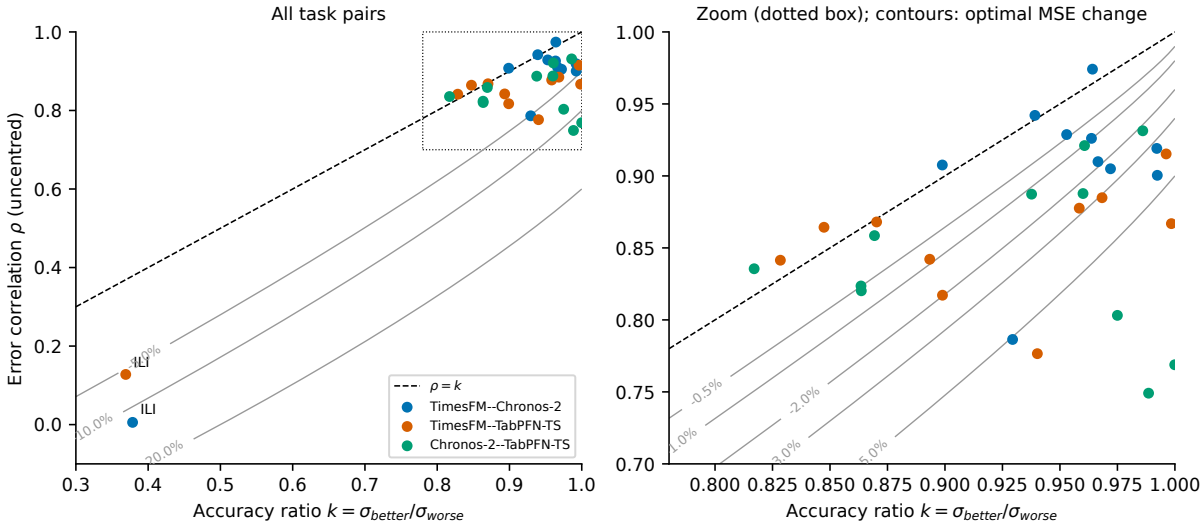


Figure 2: Task pairs on the correlation–accuracy plane with Bates–Granger iso-gain curves.

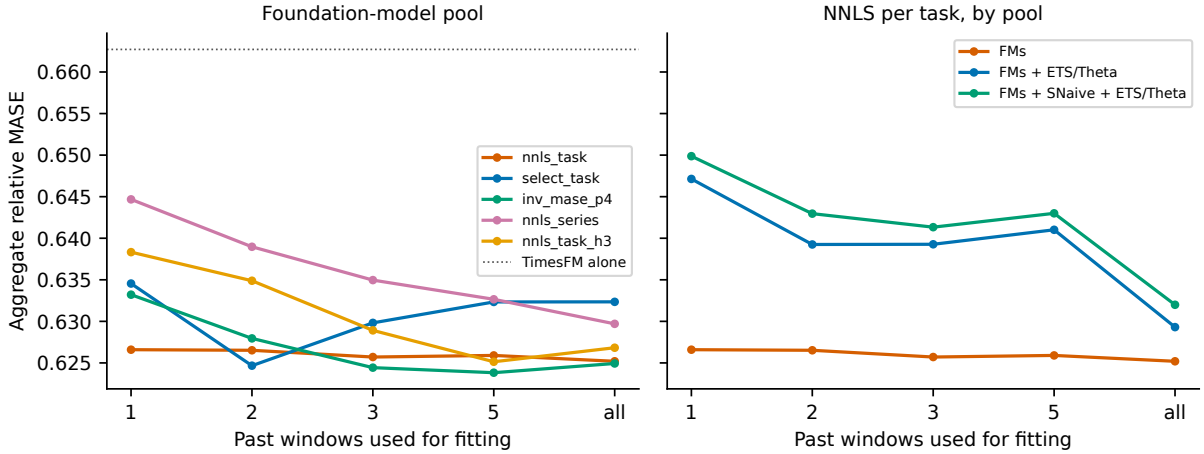


Figure 3: Combiner learning curves over the number of past windows used for fitting.

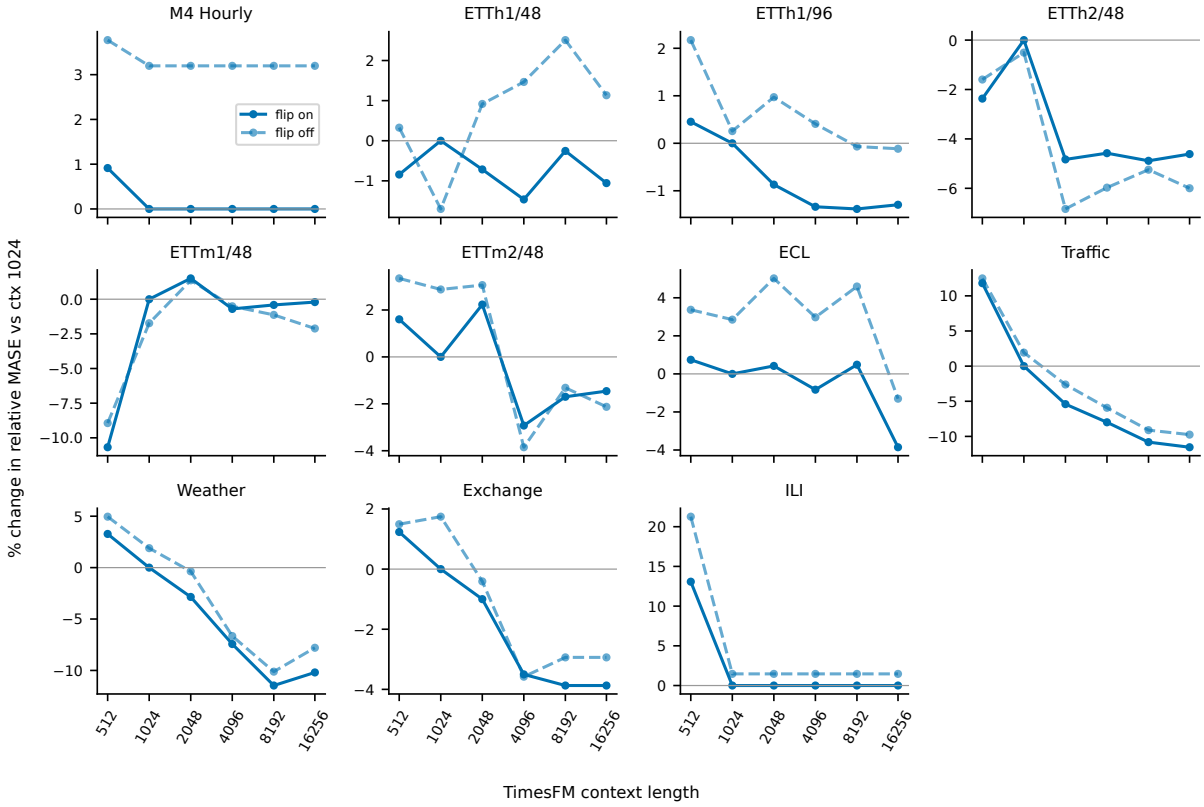


Figure 4: TimesFM context-length curves per task (% change vs ctx 1024), flip invariance on and off.

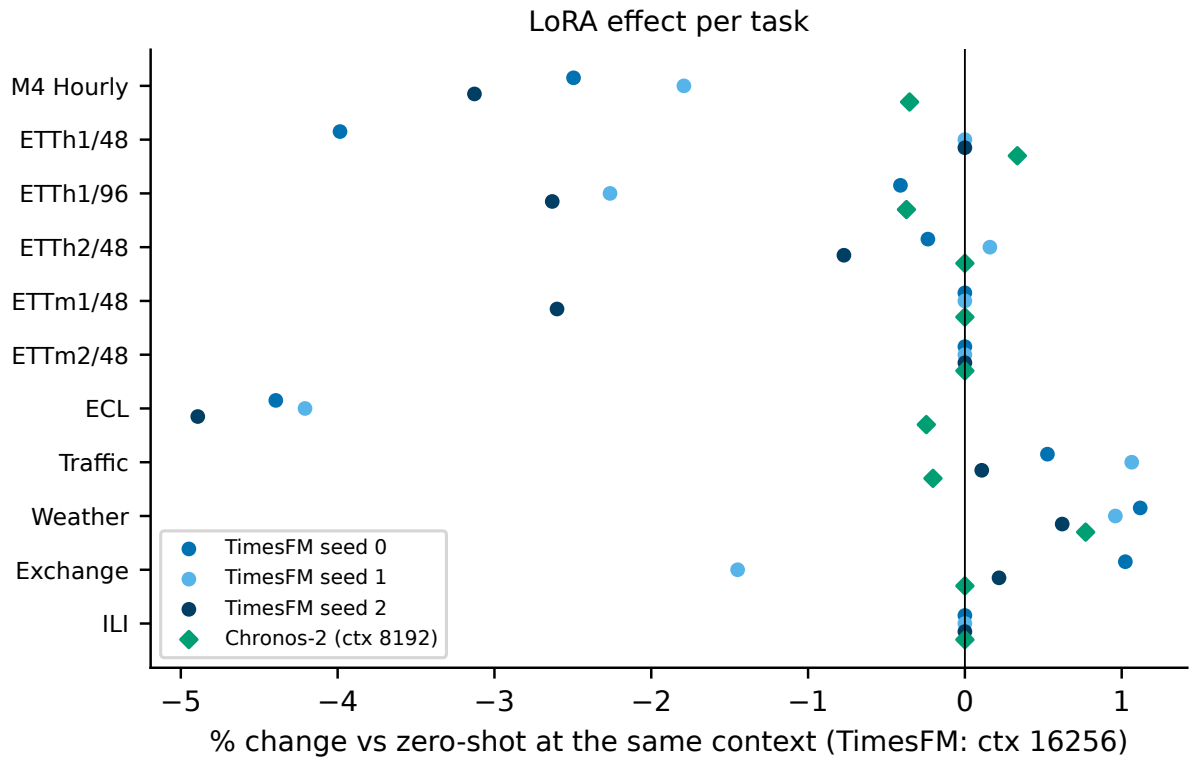


Figure 5: LoRA effect per task: three TimesFM seeds and Chronos-2.

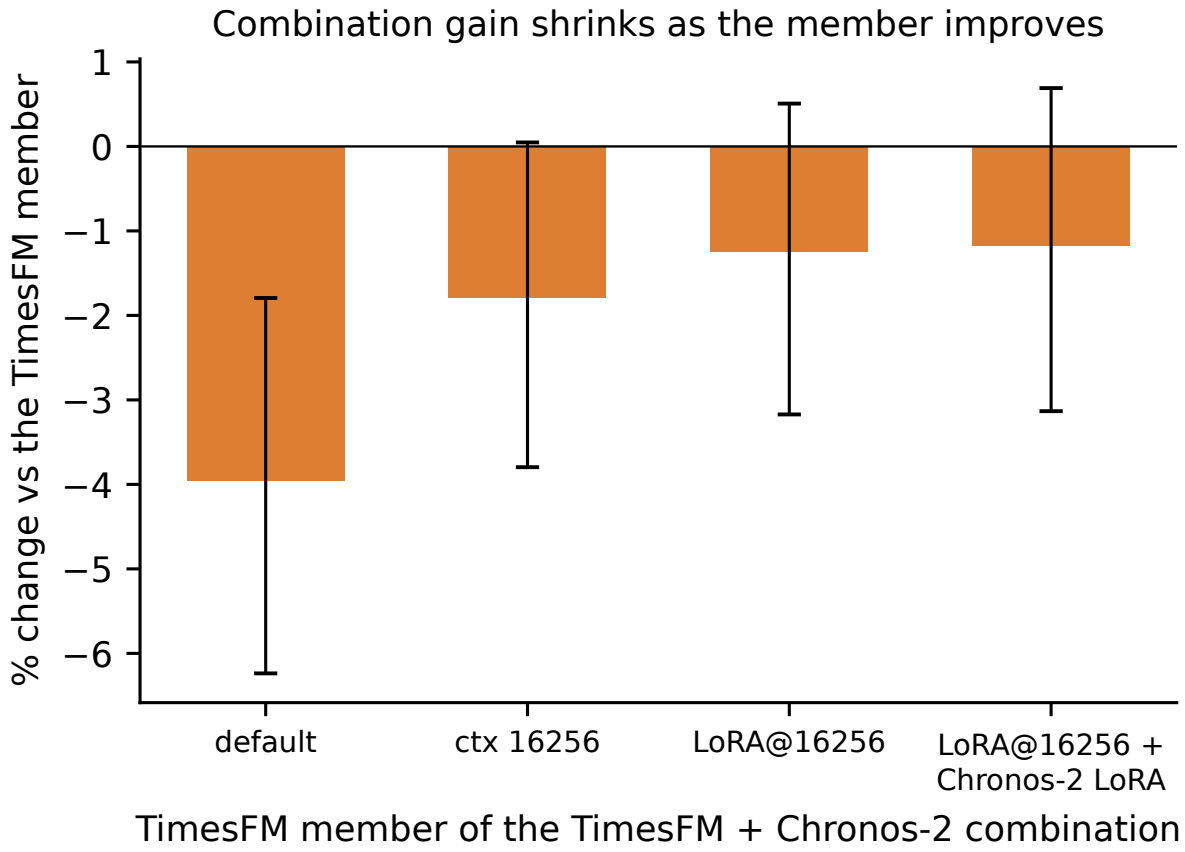


Figure 6: Combination gain with 95% intervals for successively improved TimesFM members.

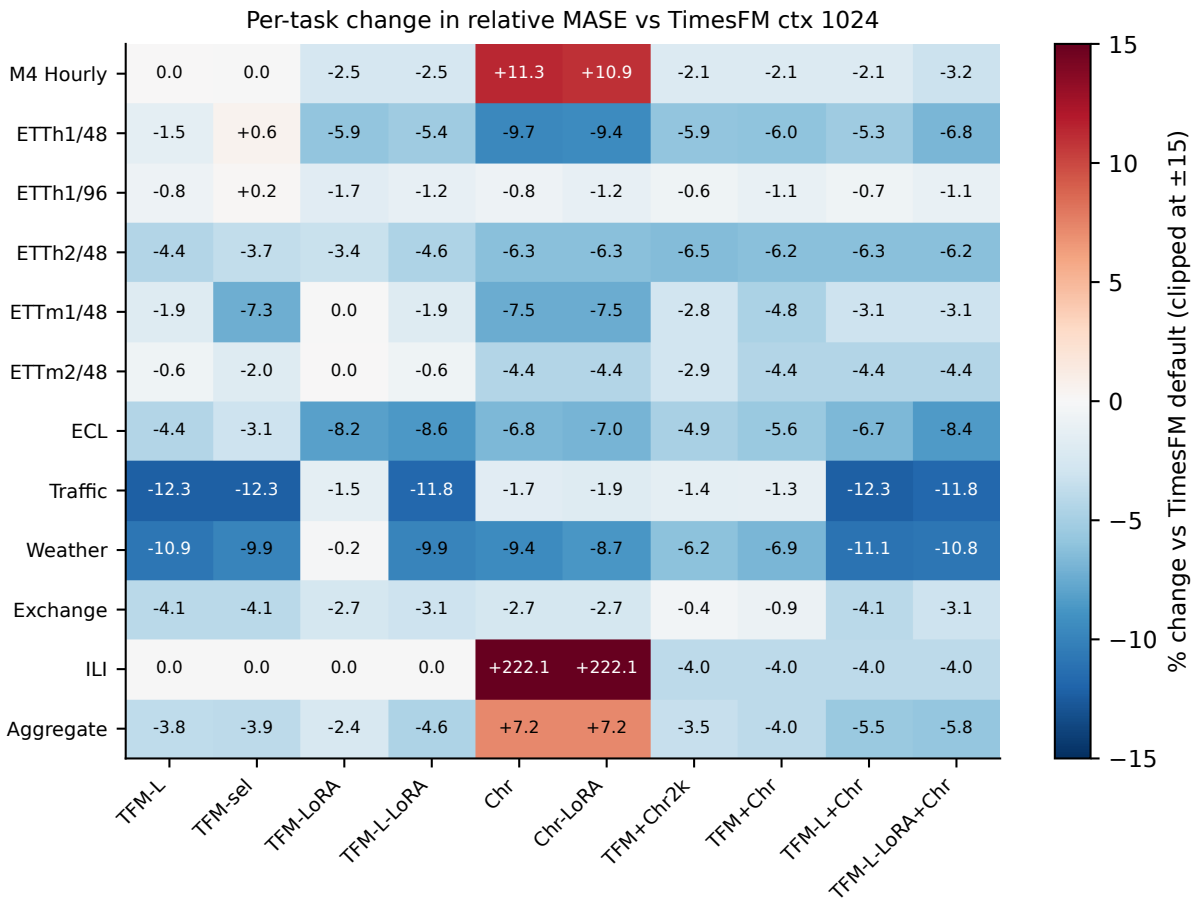


Figure 7: Per-task change vs default TimesFM for the main strategies (common evaluation set).

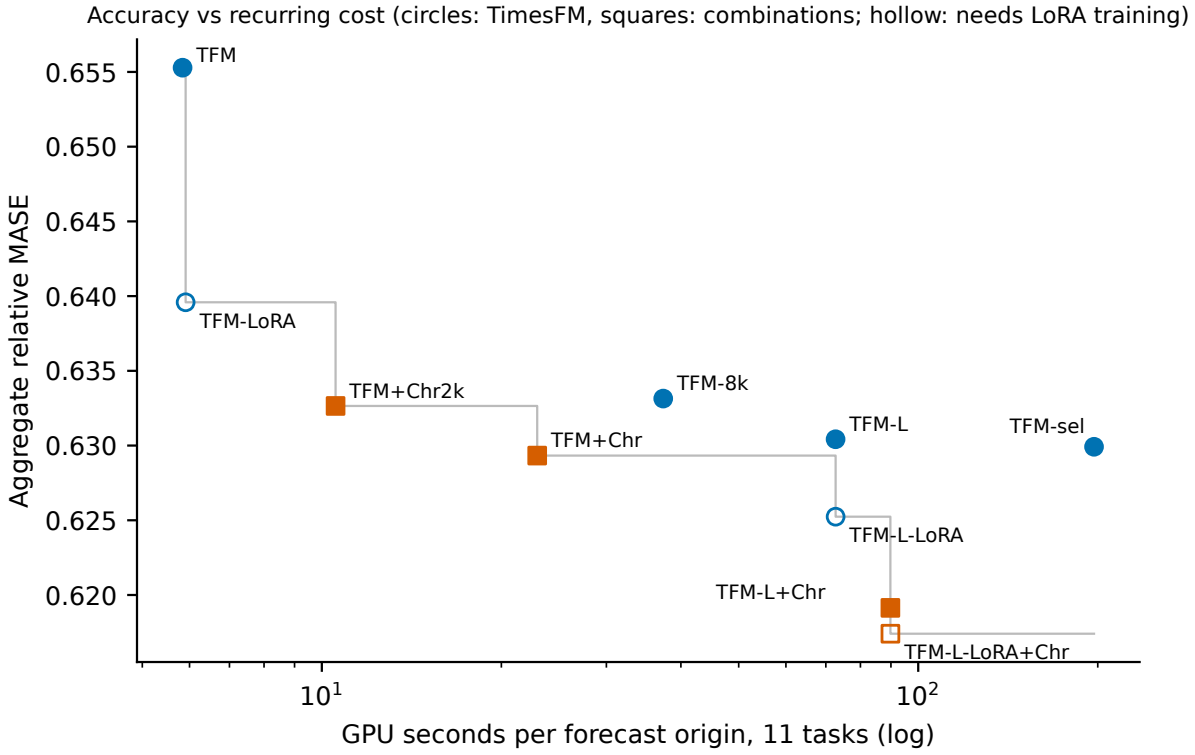


Figure 8: Accuracy vs recurring inference cost; the grey step is the cost–accuracy frontier. Chronos-2 alone (0.702, 17 s) and TabPFN-TS (0.758, ≈4,100 s) lie outside the plotted range.